Table 1: Detection and fairness performance (%) on **FairForge-2** (public SELECT/validation, seed 17), trained on FairForge-2 public TRAIN. Fairness is computed over the 8 sex × age (<30/30+) × race (White/Non-White) intersectional subgroups, as in FairForensics (arXiv 2608.01661) Table III. Lower fairness scores indicate better demographic fairness; higher AUC means better detection. **Bold** and underline denote the best and second-best results. All rows share one architecture (Xception, 20.81M parameters), so the comparison isolates the training objective. These are validation, not TEST, scores; no result here has been evaluated on TEST.

Model Type	Detector	$F_{\text{FPR}}\downarrow$	$F_{\text{MEO}}\downarrow$	$F_{\text{DP}}\downarrow$	$F_{\text{OAE}}\downarrow$	AUC↑	Params (M)
CNN-based (Xception)	Ori (BCE)	39.90	15.23	4.29	0.82	99.80	20.81
Fair-based	DAG-FDD WACV'24	40.43	14.16	4.19	<u>0.82</u>	<u>99.81</u>	20.81
	DAW-FDD WACV'24	46.41	16.79	4.15	0.84	**99.84**	20.81
	Ding et al. CVPR'26 (corrected GDA, our run)	**18.83**	**8.62**	5.12	2.19	99.06	20.81
Ours (CES-FL, head ablation)	CES-FL P: original-only BCE	38.23	14.69	**4.00**	0.99	99.81	20.81
	CES-FL A: weighted BCE	47.38	20.06	4.13	1.10	99.81	20.81
	CES-FL B: A + soft error disparity	39.56	14.78	<u>4.02</u>	**0.68**	99.81	20.81
	L_A: plain-head BCE (matched control)	42.70	21.24	4.52	0.86	99.81	20.81
	L_H-PS: hierarchical-real + probe	53.78	21.24	4.43	1.00	99.75	20.81
	RF-S: real-only FPR equalisation	41.09	21.06	4.52	0.93	99.80	20.81
	RF-SJg: RF-S + gentle JPEG aug	<u>35.67</u>	<u>9.98</u>	4.31	1.08	99.75	20.81